

Introduction to data science for wind energy systems

Exercise: Large-scale wind generation time series

Data for the exercise are given in four files (in subfolder “Data”):

- “Wind_generation_simulated.csv”: One year of simulated onshore wind generation data for 3 countries
- “Solar_generation_simulated.csv”: One year of simulated solar generation data for the same 3 countries
- “Wind_generation_simulated_model2.csv”: One year of simulated onshore wind generation data (only 1 country), using a different simulation model (model 2)
- “Wind_generation_measured.csv”: One year of measured onshore wind generation data (only 1 country)

All data are given in GMT time zone. All data are given as standardized generation, i.e., as share of installed capacity (0 means no generation and 1 means generation at full installed capacity).

Exercise 1: Wind generation in multiple countries

Load the simulated wind generation data for the three countries.

1.1: Visualize the data as time series, on hourly resolution, daily resolution (daily mean) and monthly resolution (monthly mean).

1.2: Visualize the data also as PDFs (hourly data). You can use different methods (e.g., histogram based or kernel density estimation). Which PDF visualization method appears the most appropriate for the data? Why?

1.3: Calculate mean, SD and relative SD (RSD) for each country (hourly data). Which country has the highest RSD? Can you think of any reason why? Which country has the lowest RSD? (RSD = SD divided by mean)

1.4: Calculate aggregate onshore wind generation of the three countries, assuming each country has the same amount of installed capacity (aggregate generation is the sum of the generation at each time step; however, note that also the aggregate generation should be given as standardized generation).

Calculate the same visualizations and descriptive statistics for the aggregate generation as for the individual countries in 1.1-1.3. How does the PDF of the aggregate generation look compared to the individual countries? Compare the RSD of the aggregate generation to the RSDs of the individual countries. How does it compare? Why?

1.5: Calculate hourly generation ramp time series for the individual countries and for the aggregate generation. Visualize the regeneration ramp PDFs for the individual countries and for the aggregate and calculate respective the ramp SDs. How do the aggregate ramps look compared to the individual countries? Why?

Exercise 2: Wind and solar generation in multiple countries

Keep the data from Exercise 1 open and load also the simulated solar generation data for the three countries.

2.1: Visualize the solar generation data as time series, on hourly resolution, daily resolution (daily mean) and monthly resolution (monthly mean). How does it look compared to wind generation? Why?

2.2: Calculate correlations (Pearson correlation) between all the wind generations. What can you say about the correlations? Why are some low and some high?

2.3: Calculate correlations (Pearson correlation) between all the solar generations. What can you say about the correlations? How do they appear compared to the wind generation correlations? Why do you think that is?

2.4: Calculate correlations (Pearson correlation) between all the wind and the solar generations. What can you say about the correlations between wind and solar generation?

2.5: Calculate aggregate onshore wind and solar generation of the three countries, assuming each country has the same amount of installed capacity for both wind and solar (give the aggregate generation as standardized generation). Compare the RSD of this aggregate wind and solar generation to the aggregate wind generation from 1.5. How does it compare?

Exercise 3: Compare and validate wind simulation models

Keep the data from Exercise 1 open and load also the simulated wind generation data using a different model (model 2). Load also the measured wind generation data.

3.1: Plot the two simulations models and the measured for Finland, on hourly resolution, daily resolution (daily mean) and monthly resolution (monthly mean). How well would you say the two models represent the measured data?

3.2: Compare the two models regarding validation to the measured data in terms of RMSE (hourly data). Which model looks better?

3.3: Compare the two models regarding validation to the measured data in terms of distributional properties (look at the PDFs, calculate EMD metrics, look at very low/high generation value likelihoods, etc.). Which model looks better?

3.4: Compare the two models regarding correlation to the measured data. Which model looks better?

3.5: Compare the two models in getting the annual production similar to the measured data. Which model looks better?